

VITYARTHI COURSE PROJECT

PROJECT TITLE: Library Management System

COURSE TITLE: [Programming in Java](#)

COURSE CODE: [CSE2006](#)

FACULTY: [Manorma Chouhan](#)

COURSE SLOT: [F11 + F12](#)

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INTRODUCTION

- Library Management System is an application that is used to manage the resources efficiently by libraries
- It is used to manage various jobs like the addition of books, the removal of books, showcasing all the books, borrowing books & returning of books
- This application will help us track how many books are available in the library & how many books were issued to the students
- It maintains the records online, which helps to store data & access it when necessary
- It provides users with easy access to library resources through the online directory
- It reduces the workload of management, as most of the manual work is reduced

PROBLEM STATEMENT

- The purpose of the Library Management system is to allow for storing details of a large number of books & allow for adding, searching, borrowing, and returning facilities
- The problem occurred before having a digital library system, includes:
 - Loss of records - Due to human error, it is possible to lose the records of the files or also due to natural disasters
 - Space-consuming - The space for physical storage of records is limited & making it hard to fetch the required record

- Cost-consuming - As physical documentation of maintaining library records requires paperwork

OBJECTIVES

- The main objective of the Library Management System is to organise & manage the library tasks
- Users can search for books & renewal of books online
- Applying the concepts from the core course, such as pandas, dataframes, functions, loops & datatypes to create the application
- Testing & validating the application with sample inputs for smooth running

SCOPE OF PROJECT

- The document only covers the required specifications of the Library Management System
- This project is basically updating the manual library system into a software-based application so that the user can check the availability of books
- After digitalising, the system is implemented, and less human effort will be required to maintain the library, thus reducing the overall cost
- To utilise resources effectively by increasing their productivity

FUNCTIONAL REQUIREMENTS

- **Admin Module:**
 - Add Books -
 - Admin can add books to the system by entering the title of the book & the author's name
 - Removal of Books -
 - Admin can remove books from the system by entering the title of the book & the author's name
- **Student Module:**
 - Student's login -
 - Students have to use their user ID and password to login
 - Show Books option -
 - Students can access this option to view the list of all books present in the library
 - Borrow a book -
 - Students can borrow a book on a particular day & mention their returning date using this option
 - Return a book -
 - Students can return the book that they have borrowed from the library
- **Data input:**
 - The system allows the user to choose their option, whether they want to add a book, remove a book, borrow a book, return a book or exit

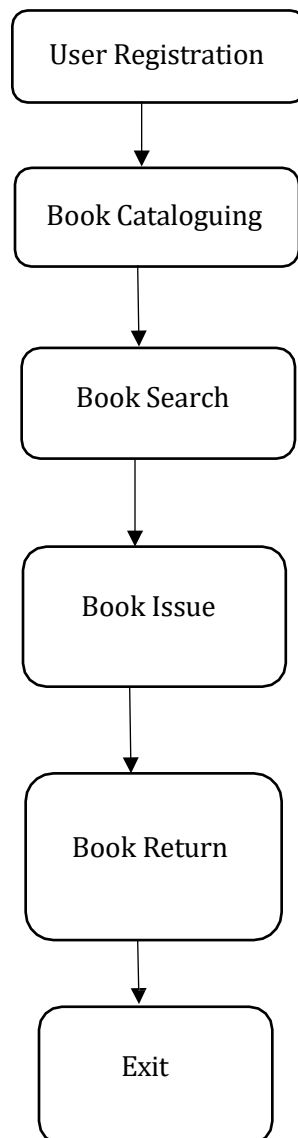
NON-FUNCTIONAL REQUIREMENTS

- Usability Requirements:
 - The user interface is interactive, simple & easy to understand
 - The system handles all the expected & non-expected errors in ways that prevent loss of information
- Security Requirements:
 - The system uses a secure database
 - Not all users can edit or modify anything except their personal information
 - Proper user authentication is provided
 - No one can hack the user's password
- Performance Requirements:
 - The system accommodates a large number of books & users without any errors
 - It flashes the information when accessed within or no longer than 5 seconds
- Software Quality Attributes:
 - The quality of the system is maintained in such a way that it can be very user-friendly to all users of the system
- Hardware Constraints:
 - The system has backup capabilities
 - The system requires a database for storing the persistent data

SYSTEM ARCHITECTURE

- The system architecture for this system defines the structural design of its components & how it interacts to provide functionalities
- The system is designed as a 3-tier architecture consisting of:
 - Presentation layer -
 - The user interface accessed by the librarian & the user for interacting with the system
 - Application layer -
 - It includes all the core operations like borrowing, returning, addition, and removal
 - Data layer -
 - The system stores all persistent data, including books & profiles

WORKFLOW DIAGRAM



DESIGN DECISIONS & RATIONALE

- Architecture: 3-tier layered design
 - Decision - Adopting 3-tier architecture (Presentation/Application/Data)
 - Rationale - Separating concerns enabling modular development & easier maintenance

- Database design with normalization:
 - Decision - Using a relational database with normalized tables for books
 - Rationale - Normalization reduces data redundancy & ensures data consistency & integrity
- Unique identification using barcodes:
 - Decision - assigning unique barcodes to books & users
 - Rationale - Barcodes enhance speed & accuracy in transactions
- Access control & role management:
 - Decision - Defines roles with specific permissions
 - Rationale - Protects system security by restricting sensitive operations

IMPLEMENTATION DETAILS

- The system can keep track of the members joining & leaving the library, borrowing & returning of the books
- Checking the availability of the books using different attributes as parameters
- Frontend: Uses HTML, CSS, JavaScript to interact with the system via web browsers
- Backend: Using languages like PHP, Python, Java or JavaScript
- Database: Relational databases like MySQL are used to store structural data about books for efficiency & integrity

TESTING APPROACH

- Unit testing:

- Each individual form of the program was tested to ensure that each performs the function for it was designed
- This testing involves the testing of the interface to ensure that the information flows into & out of the program unit under task
- The testing involves the data validation for each form & the complete functionality of the form itself
- Integration testing:
 - The integration testing was performed on the unit test modules
 - These modules were combined into subsystems, which were then tested
 - The goal was to check whether the integration of the modules was poor & to detect design errors by focusing on testing the interconnection between modules
- System testing:
 - The entire software system was tested
 - The goal was to see if the software met its requirements
 - The functionality along with the security, recovery etc. features of the system is also tested
 - System tests are designed to validate a fully developed system with a view of assuming that it meets its requirements
- User acceptance testing:
 - The users of the system with the inputs of realistic data performed this testing
 - The testing is done in controlled environment with logging of all errors

- Based on the errors found in the system, the users accepted the system with few modifications to the system, in order to rectify the errors

SCREENSHOTS/RESULTS

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\vityarthi_project> javac LibraryManagement.java
PS C:\vityarthi_project> java LibraryManagement
===== WELCOME TO VIT-BHOPAL LIBRARY MANAGEMENT =====

===== CREATE ACCOUNT =====
Enter your name: Nivetaa T
Enter your User ID: 112007

ACCOUNT CREATED SUCCESSFULLY!
Name      : Nivetaa T
User ID   : 112007

===== VIT-BHOPAL LIBRARY MANAGEMENT =====
1. Add Book
2. Show Available Books
3. Remove Book
4. Borrow Book
5. Return Book
6. Exit

Enter Your Choice: |
```

```
===== VIT-BHOPAL LIBRARY MANAGEMENT =====
1. Add Book
2. Show Available Books
3. Remove Book
4. Borrow Book
5. Return Book
6. Exit

Enter Your Choice: 1

===== ADD NEW BOOK =====
Enter Book Name: JAVA
Enter Author Name: JAMES GOSLING

BOOK ADDED SUCCESSFULLY!
Book ID: 109
```

===== VIT-BHOPAL LIBRARY MANAGEMENT =====

1. Add Book
2. Show Available Books
3. Remove Book
4. Borrow Book
5. Return Book
6. Exit

Enter Your Choice: 2

===== AVAILABLE BOOKS =====

ID	Book Name	Author	Status
101	CALCULAS	B.S. GREWEL	Available
102	COMP_CHEMISTRY	ALEXANDAR	Available
103	EEE	BATTACHARYA	Available
104	EFFECTIVE COMMUNICATION	G.V.PRAKASH	Available
105	THE_DOWNTOWN_CITY	GERONIMO	Available
106	THE GREAT BRITAN	RAMCHARAN	Available
107	THE LION KING	HIT MAN	Available
108	THE TALE OF GALLET	VK DAS	Available
109	JAVA	JAMES GOSLING	Available

===== VIT-BHOPAL LIBRARY MANAGEMENT =====

1. Add Book
2. Show Available Books
3. Remove Book
4. Borrow Book
5. Return Book
6. Exit

Enter Your Choice: 3

===== VIT-BHOPAL LIBRARY MANAGEMENT =====

1. Add Book
2. Show Available Books
3. Remove Book
4. Borrow Book
5. Return Book
6. Exit

Enter Your Choice: 4

===== BORROW BOOK =====

Enter Book Name: THE TALE OF GALLET

YOU BORROWED THE BOOK SUCCESSFULLY!
Book: THE TALE OF GALLET

```
===== VIT-BHOPAL LIBRARY MANAGEMENT =====
```

1. Add Book
2. Show Available Books
3. Remove Book
4. Borrow Book
5. Return Book
6. Exit

```
Enter Your Choice: 5
```

```
===== RETURN BOOK =====
```

```
Enter Book Name: THE TALE OF GALLET
```

```
BOOK RETURNED SUCCESSFULLY!
```

```
===== VIT-BHOPAL LIBRARY MANAGEMENT =====
```

1. Add Book
2. Show Available Books
3. Remove Book
4. Borrow Book
5. Return Book
6. Exit

```
Enter Your Choice: 6
```

```
Thank you for using VIT-Bhopal Library Management System!
```

CHALLENGES FACED

- User interface:
 - Designing a simple interface where all non-technical users can handle was indeed a challenge
 - Providing automatic navigation for searching, issuing, returning books
- Technical Implementation:
 - Choosing the right technology stack was hard and time consuming
 - Dealing with network issues
- Initial data entry was incorrect which did make my process longer

KEY TAKEAWAYS

- It improved my understanding of the database design
- It provides the hands-on experience with creating, reading, updating & deleting operations for managing records
- It made me understand how libraries function
- It provides better knowledge of the system workflow
- It provides an exposure to the real-world constraints

FUTURE ENHANCEMENTS

- There is a future scope of this facility that many more features like tutorials can be added
- A feature of group chat where students can discuss about their issues, concerns & doubts and clarify immediately
- This makes it more interactive & more user friendly

REFERENCES

- Programming in Java course from the VITYARTHI portal
- Core concepts from the course "Programming in Java"
- Google for learning regarding documentation
- Feedback from:
 - Faculty - Manorma Chouhan

